SAIence NSX-1

NSX-1 Scene Recall Help

Inputs, Output, Parameter and Command of the SIMPL+ Module

Item	Description
Product	NSX-1 lighting control system (lighting modules connected by RS485)
Software	SIMPL+ module `NSX-1 Scene Recall.usp` (3-Series and 4-Series) SIMPL Windows
Audience	Programmers of Crestron control processors
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1. Scope of This Document

This document describes the inputs, output and parameter of the SIMPL+ module NSX-1 Scene Recall, and the command it sends. The module receives digital signals such as touch panel buttons, builds the command that recalls a scene for an NSX-1 group, and outputs it to a COM port.

- Names and functions of the symbol
- Format of the command
- Use in a SIMPL Windows program
- Reference Name and trace output

Group and scene settings of the NSX-1 lighting modules, and v commands (on, off, dimming), are not covered.

1.1 Terms

Term	Meaning
Command	Character string sent on the NSX-1 line. This module sends a 12-character scene recall command.
Group number	Number of a set of lighting circuits (1 to 999). 3 digits in the command.
Scene number	Number of a combination of light levels for a group. 4 digits in the command. The levels are stored in the lighting modules.
Reference Name	Field that SIMPL Windows adds to every SIMPL+ symbol. A name can be given to each placed symbol.
Trace	Message written to the debugger with the SIMPL+ function `Trace`.

2. Names and Functions of the Symbol

Figure 2-1 shows the symbol of this module placed in a SIMPL Windows program.

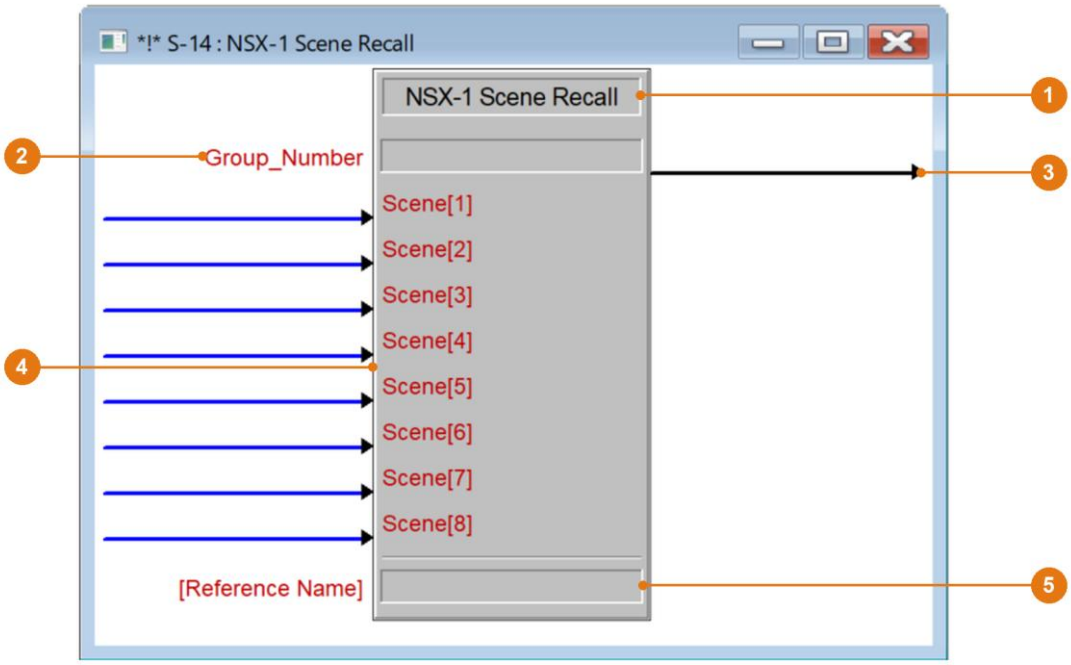


Figure 2-1 NSX-1 Scene Recall symbol

No.	Name	Type	Function
1	NSX-1 Scene Recall	Symbol name	Module name set by `#SYMBOL_NAME`.
2	Group_Number	Parameter (integer)	Group number placed in the command. Enter 1 to 999 as a decimal value such as `10d`. There is no default value, so the field shows the parameter name until a value is entered.
3	TX$	Output (serial)	The built command. Connect it to `TX$` of the COM port symbol.
4	Scene[1] to Scene[8]	Inputs (digital)	A rising edge (press) sends the command once. Scene[1] is scene 0 and Scene[8] is scene 7.
5	Reference Name	SIMPL Windows field	Name of the symbol, shown at the start of the trace message. It may be left blank (Chapter 5).

Note

SIMPL Windows draws the parameter field on the same row as the first input. The module declares `DIGITAL_INPUT _SKIP_;` before the inputs to keep the first row empty, so the Group_Number field does not cover Scene[1].

3. Command Sent by the Module

3.1 Command Format

The command has the following 12 characters. No terminator such as CR or LF is added.

Position	Characters	Content
1	`g`	Indicates a command addressed to a group
2 to 4	`010`	Group number (3 digits, zero-padded). Value of Group_Number
5 to 8	`sccc`	Scene recall instruction
9 to 12	`0001`	Scene number (4 digits, zero-padded). Number of the pressed input minus 1

The command is built by the SIMPL+ function `MakeString` with the format `"g%03dsccc%04d"`.

3.2 Command Examples

Group_Number	Pressed input	Command sent	Result
10d	Scene[1]	`g010sccc0000`	Group 010 recalls scene 0
10d	Scene[2]	`g010sccc0001`	Group 010 recalls scene 1
10d	Scene[3]	`g010sccc0002`	Group 010 recalls scene 2
20d	Scene[8]	`g020sccc0007`	Group 020 recalls scene 7

> **Note**
> The lighting modules do not reply to a scene command, and this module does not wait for a reply.

4. Use in a SIMPL Windows Program

4.1 Preparing the Module

1. Put `NSX-1 Scene Recall.usp` and `NSX-1 Scene Recall.pdf` in the user SIMPL+ folder of SIMPL Windows.
2. Open `NSX-1 Scene Recall.usp` in SIMPL+, select 3-Series and 4-Series in Build > Control System Target, and compile.
3. Check that the time stamp of `NSX-1 Scene Recall.ush` has been updated. The module appears under User SIMPL+ in the SIMPL Windows symbol library.

4.2 Placing and Connecting the Symbol

1. Place NSX-1 Scene Recall in the SIMPL Windows program, one symbol per group.
2. Enter the group number in the Group_Number field (for example `10d`).
3. Connect digital signals such as touch panel buttons to Scene[1] to Scene[8]. Unused inputs may be left unconnected.
4. Connect TX$ to `TX$` of the COM port connected to the NSX-1 line. The TX$ outputs of several symbols may use the same signal name.
5. If required, enter a name such as a room name in Reference Name (Chapter 5).

4.3 COM Port Settings

Item	Setting
Protocol	RS485
Baud rate	9600 bps
Data bits, parity, stop bits	8, none, 1
Wiring (example: COM1 of CP4)	RTS = + (TX+/RX+), TX = - (TX-/RX-), G = GND. RX and CTS are not used.

5. Reference Name and Trace Output

Reference Name is a field that SIMPL Windows adds below every SIMPL+ symbol. A name can be given to each placed symbol, and SIMPL+ code reads it with `GetSymbolReferenceName()`. Each time this module sends a command, it writes the name and the command to the trace.

```
Trace("%s: %s", GetSymbolReferenceName(), sCmd);
```

Reference Name	Pressed input (Group_Number = 10d)	Trace message
Living	Scene[2]	`Living: g010sccc0001`
(blank)	Scene[2]	`: g010sccc0001`

When several instances of the module are placed, the debugger output shows which symbol sent each command. A blank Reference Name does not change the command or the operation.

6. Notes and Known Issues

Item	Description
Compile target	For 3-Series and 4-Series. X-Generation (2-Series) does not support `INTEGER_PARAMETER` and similar declarations, so compiling for it fails.
Group_Number	There is no default value. Enter a value for each placed symbol. Range: 1 to 999.
Scene number range	One symbol recalls scenes 0 to 7. For other scene numbers, change `MAX_SCENE` or send the command by another method.
Symbol rows	After changing the number of inputs or parameters in SIMPL+, compile and place the symbol again in SIMPL Windows. An existing instance keeps the old row positions.
Line endings of the source	Save the `.usp` file with CRLF line endings. A file with LF only fails to compile (Line 1 Missing }).
Reply	The lighting modules do not reply to a scene command. Confirm the result by the state of the lights.

Appendix A SIMPL+ Code

Full text of `NSX-1 Scene Recall.usp`.

```
/*******************************************************************************************
  SIMPL+ Module Information
*******************************************************************************************/
/*
Dealer Name:   SAIence
System Name:   NSX-1
Programmer:
Comments:      Builds the NSX-1 scene recall command g(3-digit group)sccc(4-digit scene number) and sends
it to TX$
               Scene[1] = scene 0, Scene[2] = scene 1 ... Scene[8] = scene 7
               Connect TX$ to the TX$ of the COM port (RS485, 9600 bps, 8N1)
*/


/*******************************************************************************************
  Compiler Directives
*******************************************************************************************/
#SYMBOL_NAME "NSX-1 Scene Recall"
#HELP_PDF_FILE "NSX-1 Scene Recall.pdf"
#DEFINE_CONSTANT MAX_SCENE 8


#DEFAULT_VOLATILE
#ENABLE_STACK_CHECKING
#ENABLE_TRACE


/*******************************************************************************************
  DIGITAL, ANALOG and SERIAL INPUTS and OUTPUTS
*******************************************************************************************/
DIGITAL_INPUT  _SKIP_;              // row 1 is used by the Group_Number parameter field
DIGITAL_INPUT  Scene[MAX_SCENE];
STRING_OUTPUT  TX$;


/*******************************************************************************************
  Parameters
*******************************************************************************************/
INTEGER_PARAMETER Group_Number;


#BEGIN_PARAMETER_PROPERTIES Group_Number
    propValidUnits = unitDecimal;
    propBounds = 1d , 999d;
    // no propDefaultValue: until a value is entered, SIMPL Windows shows the name Group_Number in this
field
    propShortDescription = "NSX-1 group number (1-999)";
#END_PARAMETER_PROPERTIES


/*******************************************************************************************
  Event Handlers
*******************************************************************************************/
PUSH Scene
{
    INTEGER nScene;
    STRING sCmd[20];
    nScene = GetLastModifiedArrayIndex() - 1;   // Scene[1] = scene 0
    MakeString(sCmd, "g%03dsccc%04d", Group_Number, nScene);
    Trace("%s: %s", GetSymbolReferenceName(), sCmd);   // Reference Name: sent command
    TX$ = sCmd;
}


/*******************************************************************************************
  Main()
*******************************************************************************************/
Function Main()
{
    WaitForInitializationComplete();
}
```